



SINA TAVAKOLI

Data Scientist ~ Data Engineer

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ABOUT ME

Data Scientist and Data Engineer with 4+ years of experience in scalable data solutions, ML pipelines, and real-time analytics. Strong foundation in ML, cloud architecture (AWS/GCP/Azure), and automation. Skilled in Python, SQL, and modern frameworks. Passionate about solving real-world problems with data.

SKILLS

Programming: Python, SQL, C++, R, Bash

Core Areas: Data Engineering, Data Science, Data Analysis, NLP, MLOps, LLM, Big Data, Cloud (AWS, GCP, Azure), ETL/ELT, Infrastructure as Code, OOP, Data Lakes, Data Warehousing

Tools & Frameworks: PyTorch, TensorFlow, Scikit-learn, Pandas, Numpy, Spark, Airflow, dbt, Docker, Terraform, Git, MLflow, Hugging Face, BigQuery

Visualization & BI: Power BI, Looker Studio, Seaborn, Matplotlib, Plotly

Architectures & Models: Transformers (BERT, RoBERTa, etc.), CNN, RNN, LSTM, YOLOv3, OpenCV

Other Skills: Data Pipelines, API Integration, Text Processing, Data Cleaning, Version Control, Data Ingestion, Dashboarding, Linux, Automation, Data Workflow Orchestration, Business Intelligence

EDUCATION

2024 – 2025	Mobility Program in Informatics Participating in a structured mobility program as part of my Master's studies, focusing on business informatics and its applications in data-driven decision-making, enterprise systems, and digital transformation.	Mannheim University(Mannheim,Germany)
2022 – 2024	Master of ICT Master's Thesis: <i>Textual Embeddings and Neural Networks for Emotion and Slur Prediction: A Comparative Study of BERT Variants</i> Engineered a classification framework using BERT-based embeddings to detect emotions and slurs in textual data.	Padova University(Padova,Italy)
2014 – 2019	Bachelor of Electrical Engineering Bachelor's Thesis: <i>Smart grid energy optimization via data analytics.</i> Designed a predictive model for optimizing energy consumption in smart grid systems.	Razi University(Kermanshah,Iran)

PROFESSIONAL EXPERIENCE

2023–2025 **Teaching Assistant** **Università degli Studi di Padova(Padova,Italy)**

- Led tutorials and lab sessions on network analysis using Python and R, designing hands-on activities that elevated student scores by 20%.
- Designed data-driven assignments using Python's data manipulation libraries, delivering a 90% student satisfaction rate through interactive learning methods.
- Provided mentorship in Python and R programming, helping students increase their modeling accuracy by an average of 15%.

2021–2022 **Data Scientist** **Program and Budget Organization(Tehran,Iran)**

- Processed and managed large datasets (up to 50 GB), achieving a 25% reduction in processing time by optimizing data structures with Pandas and NumPy.
- Engineered predictive models for budget forecasting using SciPy and StatsModels, enhancing forecast accuracy by 30%.

2018–2021 **Python Developer** **Fara Sanat(Teharn,Iran)**

- Contributed to Python applications for customer satisfaction analysis, adding key functionalities to improve project outcomes.
- Executed unit testing and code reviews using PyTest, decreasing bug occurrences by 20%.
- Built scalable Python applications, increasing efficiency by 50% using advanced algorithms.

PROJECTS

- **Retail Data Pipeline Automation** ([GitHub Link](#))
Built a cloud-based ETL pipeline using Airflow, PySpark, and Terraform on GCP to process retail data and visualize insights in Looker Studio.%.
- **Energy Forecasting with Data Lake** ([GitHub Link](#))
Developed an end-to-end data pipeline for energy forecasting with Airflow, PySpark, dbt, and Prophet, deployed on GCP with Power BI dashboards.%.
- **Reddit Popularity Classification Pipeline** ([GitHub Link](#))
Engineered a multi-model NLP pipeline using BERT embeddings, CNNs, and Autoencoders to classify Reddit post popularity. Integrated Hugging Face Transformers, TensorFlow, and SMOTE for embedding, modeling, and class balancing, with performance visualized via precision-recall metrics.%.

LANGUAGES

English - C1, **Italian** - A1, **German** - A2, **Persian** - Native